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Brief paper

Dynamic allocation for input redundant control systems☆

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ABSTRACT

In this paper we address control systems with redundant actuators and characterize the concepts of weak and strong input redundancy. Based on this characterization, we propose a dynamic augmentation to a control scheme which performs the plant input allocation with the goal of employing each actuator in a suitable way, based on its magnitude and rate limits. The proposed theory is first developed for redundant plants without saturation and then extended to the case of magnitude saturation first and of magnitude and rate saturation next. Several simulation examples illustrate the proposed technique and show its advantages for practical application.

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1. Introduction

Many modern control applications are characterized by the presence of multiple actuators inducing the same effects on the plant dynamics (state) or at least the same steady-state effect on the plant output. In any such situation, it is necessary to design a control system which performs a suitable allocation of the actuators, for example with the goal in mind of minimizing the total energy required by the actuators, or maybe of promoting certain actuators over the others for safety, power consumption or other application specific reasons. This context generalizes the typical situation experienced, e.g., in dual stage actuator control where a fast and small actuator is connected in parallel to a slow and large actuator. Examples of emerging control applications where control allocation is relevant are the novel actuator devices recently proposed in Huston (2005). Another important area where these solutions are needed is that of nuclear fusion reactions. In this context, the demand for fast and efficient input allocators will become more and more relevant as superconducting coils (associated with severe rate limitations) will be adopted in conjunction with conducting coils (characterized by severe magnitude limitations). For example in Walker, Johnson, Leuer, and Penaflor (2005) a relevant case study is described, where actuator redundancy handling is required. Furthermore, in Ambrosino, Ariola, Pironti, Portone, and Walker (2001) the authors adopt anti-windup schemes to solve the saturation problem during the transients. However, they specify how sometimes disturbances cause steady-state saturation which cannot be effectively managed by anti-windup schemes. Dynamic input allocation would be an effective solution to this problem.

In the literature control allocation problems have been mostly addressed with reference to specific applications. Perhaps the most studied one is reconfigurable flight control where several techniques have been proposed (see Oppenheimer, Doman, and Bolender (2006) for a comprehensive survey). Much work has been also published for control allocation of ships and underwater vehicles (see Fossen and Johansen (2006) for a survey). Finally, a relevant field where a simpler control allocation problem has been addressed in different ways is that of dual stage actuator control in hard disk drives (see Mamun, Mareels, Lee, and Tay (2003) for a survey). Even though the literature in these fields is quite disconnected, there is an increasing effort in trying to unify the different approaches (see, e.g., the recent invited session Serrani and Bolender (2006)). Most of the allocation techniques adopted in practice correspond to static allocators which aim at optimizing certain criteria on the allocated input (see, for example Harkegard and Glad (2005) and references therein). For the aerospace applications, Boskovic and Mehra (2002) recognizes the strong coupling between the control allocation problem and the presence of rate and magnitude saturation limits characterizing each actuator. Also Doman and Sparks (2002) proposes allocation techniques for nonlinear actuators, while in Petersen and Bodson (2003) efficient algorithms are proposed for the on-line computation of the optimal static allocation. Finally, in Servidia and Pena (2005) spacecraft thruster configuration is addressed based on efficient optimization methods. While static

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techniques are more popular for input allocation, some dynamic approaches have also been proposed, which are more similar to the key ideas in this paper. In Johansen (2004) a dynamic approach is proposed where the dynamical allocation asymptotically recovers the pseudo-inverse allocation for redundant linear systems. The approach is therein formulated for nonlinear systems and also applied to over-actuated ship maneuvering. The difference with respect to this paper is that we explicitly focus here on the presence of (rate and magnitude) actuator saturations and optimize the allocation in view of those saturations, whereas the approach in Johansen (2004) is much more general and leads to less intuitive and more computationally expensive solutions. A so-called "dynamic allocation scheme" was also proposed in Luo, Serrani, Yurkovich, Doman, and Oppenheimer (2005) based on a sampled-data scheme using RHC ideas. However in this last paper the term "dynamic" denotes the fact that actuator dynamics are accounted for in the allocation scheme, which is a Receding Horizon law (the solution to a static optimization problem). Conversely, in this paper we use dynamics inside the allocator equations.

The core idea of this paper is that when input redundancy is at hand, it is possible to inject an arbitrary signal in certain input directions without affecting the state response of the plant (in the strongly redundant case) or the steady-state output response of the plant (in the weakly redundant case). These signals are selected as the outputs of a suitable number of integrators whose state is adjusted on-line based on certain gains intuitively chosen to promote/penalize the different actuators based on their rate or magnitude saturation levels. The proposed ideas are presented for the linear case but they can be generalized to the nonlinear case: the generalization to nonlinear controllers is straightforward, while the generalization to nonlinear plants is more involved because closed-form solutions to the stabilization of rate and magnitude saturated plants are not readily available. In Section 2 we give some basic definitions. In Section 3 we describe the input allocation scheme for linear control systems. In Section 4 we generalize it to the case of input magnitude saturation and in Section 5 we further extend it to the case of magnitude and rate saturation. Several examples are given throughout the paper to illustrate the proposed techniques.

2. Input redundant systems

Consider a linear control system involving a linear plant

$$\begin{aligned}\dot{x} &= Ax + Bu + B_d d\\ y &= Cx + Du + D_d d,\end{aligned} \tag{1}$$

(where $x \in \mathbb{R}^n$ is the plant state, $u \in \mathbb{R}^{n_u}$ is the plant input, $y \in \mathbb{R}^{n_y}$ is the plant output and $d \in \mathbb{R}^{n_d}$ is a disturbance input) satisfying an input redundancy assumption, as formalized in the next definition.

Definition 1. The plant (1) is input redundant if one of the following two conditions is satisfied

- it is *strongly input redundant* from u if it satisfies $\mathrm{Ker}\left(\begin{bmatrix}B\\D\end{bmatrix}\right) \neq \emptyset$; in this case it is useful to introduce the following full column-rank matrix:
$$B_\perp \text{ such that } \mathrm{Im}(B_\perp) = \mathrm{Ker}\left(\begin{bmatrix}B\\D\end{bmatrix}\right); \tag{2}$$
- it is *weakly input redundant* from u to y if $P^\star := \lim_{s\to 0}(C(sI - A)^{-1}B + D)$ is a finite matrix which satisfies $\mathrm{Ker}(P^\star) \neq \emptyset$; in this case it is useful to introduce the following full column-rank matrix:
$$B_\perp \text{ such that } \mathrm{Im}(B_\perp) = \mathrm{Ker}(P^\star). \tag{3}$$

Remark 1. When there is at least one pole at the origin, the DC gain of the corresponding input–output pair is infinity so that at the steady state the corresponding input must be zero. This fact makes it impossible to re-allocate the input which must evidently remain at zero in any steady-state situation. Therefore, the requirement that $P^\star$ is finite in the weak input redundancy part of Definition 1 is not restrictive. One may also focus on a restricted number of inputs to obtain $P^\star < \infty$. ○

In this paper we address the problem of exploiting the input redundancy of the plant (1) to induce desirable input allocations during the evolution of a generic control system which can be thought of as the feedback interconnection of (1) with the following linear controller[1]

$$\begin{aligned}\dot{x}_c &= A_c x_c + B_c u_c + B_r r\\ y_c &= C_c x_c + D_c u_c + D_r r,\end{aligned} \tag{4a}$$

$$u_c = y. \tag{4b}$$

In this paper, an input allocator block is inserted between the redundant input u of the plant and the controller output y_c. In particular, for the case of strong input redundancy, the input allocator will be designed in such a way that it is completely invisible to the controller. Static allocation would do a similar job in this case. Conversely, with weak input redundancy, the allocator will cause transients at the plant output and will inevitably interact with the control system, so that its speed must be sufficiently slow. For this last case, static allocators cannot be used and the theory developed here provides effective solutions to this problem.

3. Linear dynamic input allocation

3.1. Strong input redundancy

Assume that the plant (1) is strongly input redundant and consider the basis of the null space of $\begin{bmatrix}B\\D\end{bmatrix}$ introduced in (2). Note that the matrix $B_\perp$ satisfies $B_\perp^{\mathrm{T}} B_\perp > 0$ and $\begin{bmatrix}B\\D\end{bmatrix} B_\perp = 0$.

Consider the addition of the following dynamic input allocator at the input of the plant (1)

$$\begin{aligned}\dot{w} &= -KB_\perp^{\mathrm{T}}\bar{W}u\\ u &= y_c + B_\perp w,\end{aligned} \tag{5}$$

where K and $\bar{W}$ are suitable matrices to be specified next. Then the following result holds.

Theorem 1. *Assume that the plant (1) is strongly input redundant and consider the control system with input allocation namely (1), (4), (5) and (2) and the control system without input allocation namely (1) and (4), $u = y_c$.*

If K and $\bar{W}$ in (5) are symmetric and satisfy $K > 0$ and $B_\perp^{\mathrm{T}}\bar{W}B_\perp > 0$, then:

(1) *the control system with input allocation is internally stable if and only if the control system without input allocation is internally stable;*
(2) *given any initial condition $(x(0), x_c(0), w(0))$ and any selection of the external signals $r(\cdot)$, $d(\cdot)$, the plant output responses of the two systems coincide for all times.*

Proof. It is straightforward to verify that since $BB_\perp = 0$, the control system with input allocation can be written as the cascade interconnection of a first subsystem consisting in (4) and (1)with $u = y_c$, which drives via the signal y_c a second subsystem consisting in the dynamic allocator (5). Therefore the plant output y is independent of the dynamic allocator output w. To show internal stability it is sufficient to show internal stability of the dynamic allocator (because of the cascade interconnection). To this end, consider the Lyapunov function $V(w) = \frac{1}{2}w^{\mathrm{T}}K^{-1}w$ and note

[1] The controller (4) can be generalized to a generic locally Lipschitz controller. The attention here is restricted to the linear case to increase the clarity of the paper.

that $\dot{V} = -w^{\mathrm{T}}B_{\perp}^{\mathrm{T}}\bar{W}B_{\perp}w - w^{\mathrm{T}}B_{\perp}^{\mathrm{T}}\bar{W}y_c$, which implies internal stability because $B_{\perp}^{\mathrm{T}}\bar{W}B_{\perp} > 0$ by assumption. •

Remark 2 (*Speed of the Dynamic Allocator*). Theorem 1 establishes that any (arbitrarily fast) dynamics in the dynamic allocator (5) can be implemented because the dynamics of w are completely invisible to the plant-controller dynamics. Due to this fact, given desirable matrices W_d and K_d, the speed of the allocator can be adjusted from extremely slow to extremely fast by selecting different values of a positive scalar ρ in the selection $K = \rho K_d$, $\bar{W} = W_d$. Indeed, by Theorem 1, any positive value of ρ will be allowed and given any controller output y_c^* the magnitude of ρ will regulate the speed of convergence to the steady-state input allocation, which is independent of K_d and is given by

$$u^{\star} = \left(I - B_{\perp}(B_{\perp}^{\mathrm{T}}\bar{W}B_{\perp})^{-1}B_{\perp}^{\mathrm{T}}\bar{W}\right)y_c^{\star}. \tag{6}$$

This equation is easily derived from (5) by imposing the steady-state condition $\dot{w} = 0$ and corresponds to the solution to the following optimization problem:

$$\begin{aligned} &\min_{w} J(u) := u^{\mathrm{T}}\bar{W}u, \text{ subject to} \\ &u = y_c^{\star} + B_{\perp}w, \end{aligned} \tag{7}$$

as a matter of fact, imposing the stationary condition $\frac{\partial J(u(w))}{\partial w} = 0$, Eq. (6) is easily determined. ○

Remark 3 (*Shifting the Allocator Center*). It is worth mentioning that an arbitrary (not necessarily constant) signal $u_0 \in \mathbb{R}^{n_u}$ can be subtracted from u at the right hand side of the input allocator (5), as well as its generalizations reported next. In this way, the distance of u from u_0 (rather than from the origin) is penalized by the allocator. This trick allows one, for example, to account for non-symmetric saturations, as well as certain desirable operating conditions for some of the plant inputs. For example, if the first equation in (5) is replaced by $\dot{w} = -KB_{\perp}^{\mathrm{T}}\bar{W}(u - \bar{u}_0)$, where $\bar{u}_0$ is constant, then the steady-state input allocation (6) shifts to

$$u^{\star} = y_c^{\star} - B_{\perp}(B_{\perp}^{\mathrm{T}}\bar{W}B_{\perp})^{-1}B_{\perp}^{\mathrm{T}}\bar{W}(y_c^{\star} - \bar{u}_0),$$

which is easily shown to be the solution to the following optimization problem, generalizing (7):

$$\begin{aligned} &\min_{w} J_{\bar{u}_0}(u) := (u - \bar{u}_0)^{\mathrm{T}}\bar{W}(u - \bar{u}_0), \text{ subject to} \\ &u = y_c^{\star} + B_{\perp}w. \quad ○ \end{aligned}$$

Remark 4 (*Selecting the Parameters K and $\bar{W}$*). Theorem 1 implies that any positive definite pair of matrices K and $\bar{W}$ in the dynamic allocator (5) will guarantee internal stability of the modified closed loop. This degree of freedom can be exploited in different ways. As an example, using a diagonal $\bar{W}$, it is possible to promote or penalize the size of the different components of the plant input by suitably selecting, respectively, smaller or larger entries in the diagonal of the matrix $\bar{W}$. Indeed, the larger an entry in $\bar{W}$, the larger will be the correction automatically enforced by the allocator when the corresponding input is nonzero. As for K, it can be used to represent the control allocator with any basis B_0 of the null space of B, as a matter of fact, given an orthonormal basis $B_{\perp}$ and any basis B_0, there always exists a coordinate change T such that $B_0T = B_{\perp}$ and selecting $\tilde{K} = TKT^{\mathrm{T}}$ and defining $\tilde{w} = Tw$, one gets the equivalent representation for (5):

$$\begin{aligned} &\dot{\tilde{w}} = -\tilde{K}B_0^{\mathrm{T}}\bar{W}u \\ &u = y_c + B_0\tilde{w}, \end{aligned} \tag{8}$$

which corresponds to selecting B_0 as any basis of the null space of B in (5). The general representation (8) is especially useful when wanting to penalize the use of certain redundant directions which can be isolated in the generic basis B_0. In that case, selecting the matrix $\tilde{K}$ as a diagonal positive definite matrix, some redundant input directions (corresponding to suitable columns of B_0) can be penalized by making the corresponding entries in $\tilde{K}$ small. ○

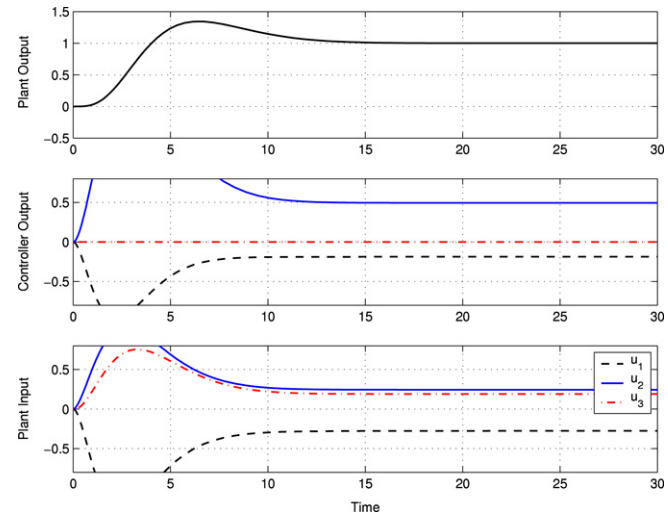

Fig. 1. Example 1: Responses to a unit step reference.

Remark 5 (*Robustness with Respect to Errors on $B_{\perp}$*). Since the proof of Theorem 1 (as well as the following theorems) relies on Lyapunov-based arguments and all the right hand sides are Lipschitz, the scheme is robust with respect to arbitrarily small errors in $B_{\perp}$. Moreover, a formal robustness analysis with nontrivial estimates of the tolerable variations of $B_{\perp}$ can be carried out based on a small gain argument. ○

Example 1. Consider the following randomly generated exponentially stable plant

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{cc|ccc} -0.157 & -0.094 & 0.87 & 0.253 & 0.743 \\ -0.416 & -0.45 & 0.39 & 0.354 & 0.65 \\ \hline 0 & 1 & 0 & 0 & 0 \end{array}\right].$$

The plant is strongly input redundant with the orthonormal selection $B_{\perp} = [-0.27512 \;\; -0.76512 \;\; 0.58215]^{\mathrm{T}}$. We design a controller guaranteeing asymptotic tracking of constant references via: (1) negative error feedback interconnection; (2) inserting an integrator; (3) designing a stabilizing LQG controller which only uses the first two input channels. The LQG controller is designed using identity matrices both for the input and state weights in the LQ index and for the covariance matrix of the state and output equation noises. The arising matrices in (4) correspond to

$$\left[\begin{array}{c|c} A_c & B_r \\ \hline C_c & D_r \end{array}\right] = \left[\begin{array}{cccc|c} -1.57 & 0.5767 & 0.822 & -0.65 & 0 \\ -0.9 & -0.501 & -0.94 & 0.802 & 0 \\ 0 & 1 & -1.61 & 1.614 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ \hline 1.81 & -1.2 & -0.46 & 0 & 0 \\ -0.62 & 1.47 & 0.89 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right],$$

and to $B_c = -B_r$, $D_c = -D_r$.

Note that this controller is purposely selected as a dull controller which never uses the third input channel, so the input allocation for a given desired plant response can be improved in terms of the size of each allocated input. We implement the input allocator (5) selecting $K = 10I$ and $\bar{W} = I$. The corresponding response to a unit step reference is reported in Fig. 1 where the top plot represents the output response y (this is independent of the control allocation), the middle plot represents the controller output y_c (here, the third input channel is zero) and the bottom plot

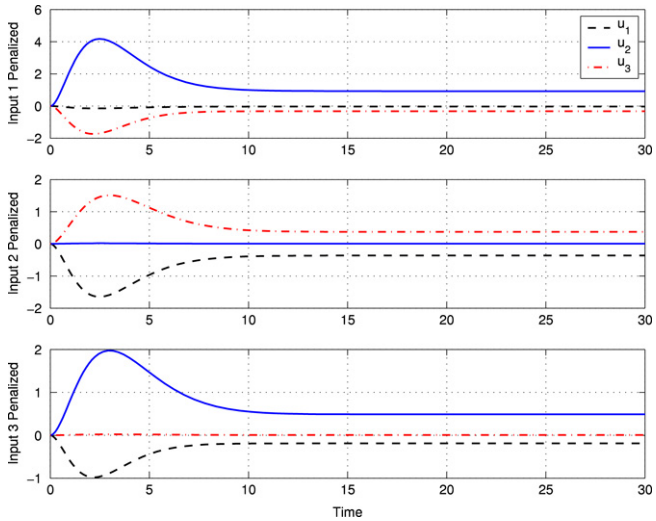

Fig. 2. Example 1: Different allocations based on the choice of $\bar{W}$.

represents the plant input u (note that the control allocator allows reducing the steady-state size of the largest input to roughly one half as compared to the middle plot).

Following the comments in Remark 4 about the selection of W, we simulate three variants of the control allocator, selecting first $\bar{W} = \text{diag}([100\ 1\ 1])$, then $\bar{W} = \text{diag}([1\ 100\ 1])$ and finally $\bar{W} = \text{diag}([1\ 1\ 100])$. The resulting input allocations are shown in the top, middle and bottom plots of Fig. 2, respectively (the plant output responses are not reported because they all coincide with the top plot of Fig. 1). Fig. 2 clearly illustrates the abilities of the control allocator to promote or penalize each control input. For this same example, an illustration of how the choice of K impacts the convergence speed of the allocator can be easily carried out by simulation (see Zaccarian (2008)).

3.2. Weak input redundancy

With weak input redundancy the dynamic allocation transients are visible at the plant output y, therefore there is a strong limit in the dynamic allocation speed commented in Remark 2. The main result for weakly input redundant plants establishes that there is a small enough speed such that the input–output allocation is effective (namely, closed-loop stability is retained). The corresponding dynamic allocator is given by (5) where $B_\perp$ now assumes the meaning in (3). In the following we will denote as "converging signal" any function $t \mapsto s(t)$ such that $\lim_{t\to\infty} s(t)$ is well defined and finite.

Theorem 2. *Assume that the plant (1) is weakly input redundant and that the control system without input allocation (namely (1) and (4), $u = y_c$) is internally stable. Consider the input allocator (5) with $B_\perp$ as in (3) and with $K = \rho\bar{K}$. If $\bar{K}$ and $\bar{W}$ are symmetric and satisfy $\bar{K} > 0$ and $B_\perp^{\mathrm{T}}\bar{W}B_\perp > 0$, then there exists a small enough ρ such that:*

(1) *the control system with input allocation (namely (1), (4), (5) and (3), $K = \rho\bar{K}$) is internally stable and*
(2) *Given any converging selection of the external signals $r(\cdot)$, $d(\cdot)$, the plant output responses of the two systems converge to the same values.*

Proof. The closed-loop system with dynamic input allocation (1), (4), (5) and (3), $K = \rho\bar{K}$ can be written in the following form

$$\begin{bmatrix}\dot{x}\\ \dot{x}_c\end{bmatrix} = A_{cl}\begin{bmatrix}x\\ x_c\end{bmatrix} + B_{cl,d}d + B_{cl,r}r + B_{cl,w}w$$

$$y_c = C_{cl}\begin{bmatrix}x\\ x_c\end{bmatrix} + D_{cl,d}d + D_{cl,r}r + D_{cl,w}w$$

$$\frac{\dot{w}}{\rho} = -\bar{K}B_\perp^{\mathrm{T}}\bar{W}(B_\perp w + y_c),$$

where the matrices with subscripts "*cl*" can be uniquely determined from the plant and controller matrices in (1) and (4). Then, by the internal stability assumption for the closed loop without input allocation, standard results on systems with two time scales (see, e.g., Khalil (1996)) can be applied to conclude internal stability of the closed loop with input allocation. As for item (2), internal stability and linearity imply that any converging selection of external inputs will lead to converging state responses. Since the plant (1) is linear, denoting by $\star$ the steady-state values of the signals, the steady-state plant output will correspond to the sum of the contributions arising from each one of the plant inputs, namely $y^\star = y^\star_{y_c} + y^\star_d + y^\star_w = y^\star_{y_c} + y^\star_d$ because $y^\star_w = P^\star B_\perp(B_\perp^{\mathrm{T}}\bar{W}B_\perp)^{-1}B_\perp^{\mathrm{T}}\bar{W}y^\star_c = 0$ by the definition of $B_\perp$ in (3). Then by uniqueness of solutions, the steady-state response $y^\star$ will necessarily coincide with the steady-state response of the closed loop without input allocation. •

Remark 6. The results in Theorem 2 imply that for a given control system where different control allocations can lead to the same steady-state output, it is possible to implement a slow enough control allocator such that the corresponding dynamics is on a different time scale from that of the plant-controller closed loop. This approach is especially useful when wanting the plant inputs to slowly drift toward a desired control allocation solution (see, e.g., the problems in Walker et al. (2005)). In this case, the value of ρ can be selected small enough so that the control allocator takes care of the slow drift which (by redundancy) is invisible to the plant output and to the feedback controller. ○

Remark 7. In the presence of both strong and weak input redundancy, it is possible to select the matrix $B_\perp$ so that the first columns are in the strong redundancy directions. Then via suitable diagonal selections of the matrix K it is possible to combine the approaches of Theorems 1 and 2 by simultaneously using different speeds of allocation. ○

Example 2. Consider the control system introduced in Example 1. The plant is evidently weakly input redundant and an orthonormal selection for $B_\perp$ is $B_\perp = \begin{bmatrix}-0.135 & 0.99 & -0.042\\ -0.56 & -0.042 & 0.825\end{bmatrix}^{\mathrm{T}}$. Control allocation can be effected by picking $K = I$ and $\bar{W} = I$ and choosing a small enough ρ to ensure closed-loop stability.

Figs. 3 and 4 illustrate the closed-loop responses with $\rho = 1$ and $\rho = 0.1$, respectively. The unstable behavior of the former case illustrates the limit on the speed of the dynamic control allocator in the case of weak input redundancy. Note that despite the very little use of u_3 in the bottom plot of Fig. 4, the dynamic allocator allows significantly reducing the magnitude of the inputs u_1 and u_2 (compare to the middle plot of the same figure). Also note that in the strong input redundancy case, as commented in Remark 2 this speed can be arbitrarily large. Finally, according to Remark 7 it is possible to make use of the strong input redundancy direction by replacing the first column of $B_\perp$ by the vector $B_\perp$ determined in Example 1. In this case the allocator speed is much more improved (it is not reported here due to space constraints).

4. Nonlinear input allocation for magnitude saturated plants

Most plants with redundant actuators are equipped with such devices to (partially) overcome the limitations of the use of a single actuator by enhancing the control capability via the use of multiple hardware, possibly characterized by different features, such as different input ranges (which can be modeled as magnitude

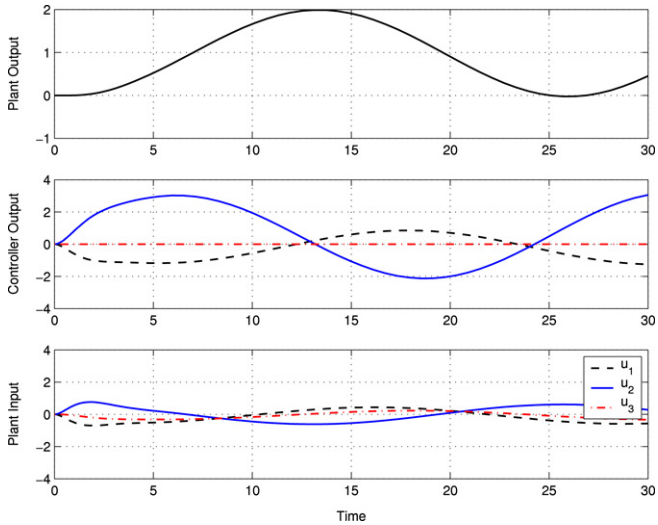

Fig. 3. Example 2: Unstable response caused by the selection $\rho = 1$.

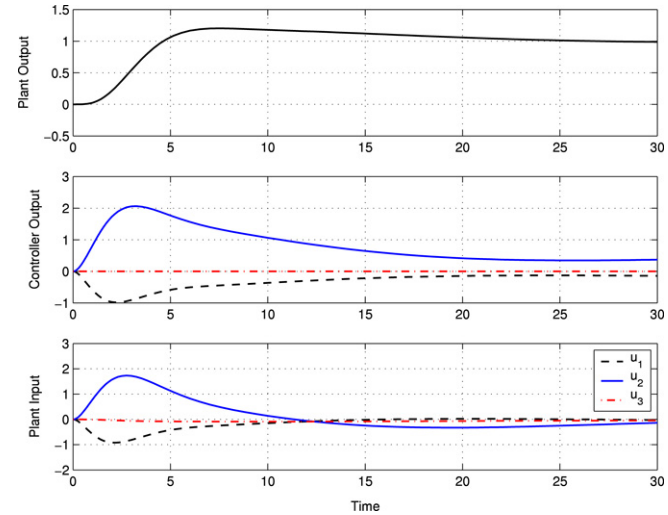

Fig. 4. Example 2: Stable response corresponding to $\rho = 0.1$.

saturation effects) and/or different input variation capabilities (which can be modeled as magnitude saturation effects). In the case when only magnitude saturation is present, the control allocation goal is mostly to promote the use of actuators capable of large excursions (namely characterized by large saturation limits) To this end, the linear allocation approach introduced in Section 3 can be suitably extended by generalizing the constant matrix $\bar{W}$ to a non-constant matrix function $W(\cdot)$ which allows adjusting online the selection of the actuators to be promoted or penalized. To be able to establish formal properties of the closed loop, such as stability and convergence, we will make use of suitable anti-windup techniques, necessary to guarantee stability recovery in light of saturation. In particular, since the control system with the allocator is nonlinear, we will use the anti-windup techniques of Teel and Kapoor (1997), Zaccarian and Teel (2002) and Forni, Galeani, and Zaccarian (2009) because they allow for nonlinear controllers. Most of the other existing anti-windup approaches cannot be adopted here because they require linearity of the controller, however if the allocator is linear, then linear anti-windup techniques could be adopted as well by considering the controller + allocator pair as an extended linear controller.[2]

To account for plant input magnitude saturation, consider the following input selection for the plant (1):

$$u = \mathrm{sat}_M(y_u + v_1) = \mathrm{sat}_M(y_c + B_\perp w + v_1), \tag{9}$$

where $y_u = y_c + B_\perp w$ is the controller output after the control allocation, v_1 is a stabilizing signal defined next and $\mathrm{sat}_M(\cdot)$ is defined as the decentralized symmetric saturation function [3] having saturation limits, respectively, $m_1, \ldots, m_{n_u}$, and where $M = [m_1 \ \cdots \ m_{n_u}]^{\mathrm{T}}$, so that for any $w \in \mathbb{R}^{n_u}$, $\mathrm{sat}_M(w) := [m_1\sigma(w_1/m_1) \ \cdots \ m_{n_u}\sigma(w_{n_u}/m_{n_u})]^{\mathrm{T}}$, with $\sigma(\cdot)$ being the unit saturation function defined for any $s \in \mathbb{R}$ as $\sigma(s) := \mathrm{sign}(s)\min\{|s|, 1\}$.

Given a small constant $\epsilon \in (0, 1)$ (for example, $\epsilon = 0.01$ has been often used in the simulation studies) the dynamic input allocator (5) is modified as follows:

$$\begin{aligned} \dot{w} &= -KB_\perp^{\mathrm{T}} W(y_u) y_u \\ y_u &= y_c + B_\perp w \\ W(y_u) &= (\mathrm{diag}((1+\epsilon)M - \mathrm{abs}(\mathrm{sat}_M(y_u))))^{-1}, \end{aligned} \tag{10}$$

where given a vector w, $\mathrm{abs}(w)$ denotes the vector whose components are respectively the absolute values of the components of w. The selection of $W(y_u)$ in (10) is especially promising because each diagonal term of $W(\cdot)$ becomes larger and larger as the argument is far from zero and approaches the respective saturation level. Also, according to Remark 2, the steady-state solution reached by (10) locally solves the optimization problem (7) with $\bar{W}$ replaced by $W(u^*)$. Therefore the steady-state allocation will be, as much as possible, far from the saturation values of each input.

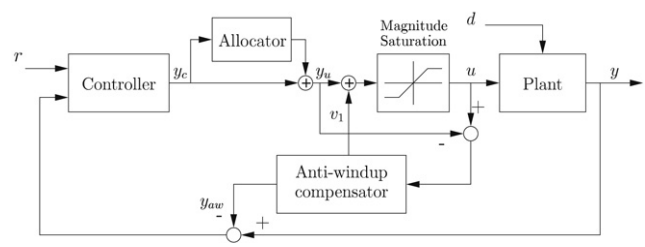

Fig. 5. Dynamic input allocation with anti-windup for magnitude saturated plants.

The allocator (10) is interconnected via Eq. (9) where v_1 is a suitable correction signal generated to guarantee closed-loop asymptotic stability in light of saturation, by the following $\mathcal{L}_2$ anti-windup compensator, which is added to the closed loop as shown in Fig. 5.

$$\begin{aligned} \dot{x}_{aw} &= Ax_{aw} + B(u - y_u) \\ y_{aw} &= Cx_{aw} + D(u - y_u) \\ v_1 &= k(x_{aw}) \end{aligned} \tag{11}$$

and interconnected to the controller dynamics (4a) by replacing the interconnection equation (4b) via the following modified interconnection:

$$u_c = y - y_{aw}. \tag{12}$$

Following the anti-windup approach of Teel and Kapoor (1997), the function $k(\cdot)$ in Eq. (11) should be selected in such a way that the system

$$\dot{x}_{aw} = Ax_{aw} + B\left(\mathrm{sat}_M(y_u + k(x_{aw})) - y_u\right) \tag{13}$$

[2] There might be specific applications where anti-windup is not necessary from a practical viewpoint, however if wanting to formally prove stability and convergence, the presence of the anti-windup compensator is mandatory.

[3] To simplify the discussion, we only focus on symmetric saturations here. However, following the ideas in Remark 3, it is possible to extend the approach to non-symmetric saturations.

is $\mathcal{L}_2$ stable from $y_u - \mathrm{sat}_{M-\varepsilon}(y_u)$ to x_{aw} (ε is an arbitrarily small constant necessary to guarantee feasibility of the construction of $k(\cdot)$). This property can be either achieved globally (whenever the plant is not exponentially unstable) or regionally (if the plant is exponentially unstable) and the corresponding selections of the function $k(\cdot)$ are nonlinear and quite involved in general, however in the special case where the plant is exponentially stable (namely, A is Hurwitz), the trivial selection $k(x_{aw}) \equiv 0$ is already sufficient for closed-loop exponential stability.

Since the focus of this paper is on the dynamic allocation aspects, rather than the anti-windup solutions, we will make use of the following definition of *feasible control signal* for (13) in the following. The more effective the selection of the anti-windup function $k(\cdot)$ (or its generalizations) is, the larger the set of feasible control signals will be. We disregard here these anti-windup aspects and rather only use the necessary stability results induced by v_1 to formalize our dynamic allocation theorems.

Definition 2. A function $y_u(t)$, $t \geq 0$ is a *feasible control signal* for (13) if $y_u - \mathrm{sat}_{M-\varepsilon}(y_u) \in \mathcal{L}_2$ and the solution of (13) satisfies $x_{aw} \in \mathcal{L}_2$

Theorem 3. *Consider an input redundant plant (1) subject to input magnitude saturation of level M and the controller (4) designed in such a way that their interconnection without saturation and without allocation (namely, $u = y_c$) is internally stable. Then, in the strong [respectively, weak] input redundancy case, the control system with input allocation and without saturation (1), (4) and (10), $u = y_c + B_\perp w$ is globally exponentially stable [respectively, it is globally exponentially stable for a small enough ρ in $K = \rho\bar{K}$]. Moreover, for any initial condition $(x(0), x_c(0), w(0))$ and input selections $r(\cdot), d(\cdot)$ such that this system generates a* feasible control signal *$y_u = y_c + B_\perp w$, the following holds for the closed loop with input allocation and with saturation (1), (4a) and (10)–(12) with $B_\perp$ as in (2) [respectively, as in (3)]:*

(1) *the plant input response u converges in the $\mathcal{L}_2$ sense to the plant input response of the control system with input allocation and without saturation;*
(2) *[if $r(\cdot)$ and $d(\cdot)$ are converging signals], the plant output response y converges in the $\mathcal{L}_2$ sense to the plant output response of the control system without saturation and without allocation.*

Proof. Global exponential stability of the closed loop without saturation (and with dynamic allocation) is easily proved by the same arguments used in Theorems 1 and 2, also noticing that for any diagonal positive definite $\bar{W}$ one gets $B_\perp^{\mathrm{T}} \bar{W} B_\perp > 0$ and that $W(\cdot)$ in (10) satisfies $W(y_u) > (\mathrm{diag}((1+\epsilon)M))^{-1} =: W_0 > 0$, so that the Lyapunov arguments of Theorems 1 and 2 apply, despite the nonlinear nature of (10).

Consider now the whole closed-loop dynamics (1), (4a), (10), (11), (9) and (12) and write them in the coordinates $(x_e, x_c, x_{aw}, w) := (x - x_{aw}, x_c, x_{aw}, w)$, which lead to a cascade structure where the first subsystem, corresponding to:

$$\begin{aligned}
\dot{x}_e &= Ax_e + By_u + B_d d \\
\dot{x}_c &= A_c x_c + B_c y_e + B_r r \\
\dot{w} &= -KB_\perp^{\mathrm{T}} W(y_u) y_u \\
y_e &= Cx_e + Dy_u + D_d d \\
y_u &= C_c x_c + D_c y_e + D_r r + B_\perp w
\end{aligned} \tag{14}$$

drives, by way of y_u, a second subsystem corresponding to (13).

Since (14) reproduces the closed loop without saturation and with dynamic allocation, then, by assumption, the signal y_u driving (13) is a feasible control signal. Hence, $x_{aw} = x - x_e \in \mathcal{L}_2$, namely x converges in the $\mathcal{L}_2$ sense to the response without saturation x_e. Therefore the response of the closed loop with saturation converges in the $\mathcal{L}_2$ sense to the response without saturation. Based on this convergence property, the proof is completed disregarding saturation and following the same steps as in the proof of Theorem 2. •

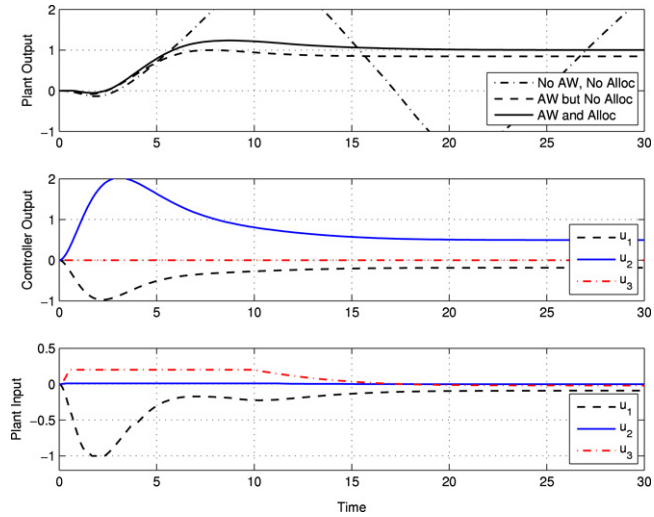

Fig. 6. Example 3: Response to the unit step reference.

Remark 8. Theorem 3 establishes that dynamic allocation can be employed for systems with magnitude saturation by way of an anti-windup compensator. The advantage in using an anti-windup approach is that one can design the control allocator to ensure stability in the absence of saturation (an easier task, in light of the actual saturation limits) and then address the saturation problem via the anti-windup action. Since effective anti-windup approaches are available, this trick allows simplifying the dynamic allocation problem for magnitude saturated systems. In particular, the theorem shows that the control allocator is effective for all the trajectories that can be handled by the anti-windup compensator. ○

Example 3. We revisit the control system introduced in Examples 1 and 2 and assume now that all three inputs are subject to saturation, with saturation limits $M = [1\ 0.01\ 0.2]^{\mathrm{T}}$. Note that these limits are randomly chosen because for any selection of M, the control allocator will automatically try to keep each control input far from its saturation limit. We consider the case where the first column of $B_\perp$ is the strongly redundant direction, while the second column completes the weakly redundant directions. However, in this case to achieve desirable responses in light of saturation, we select $K = \begin{bmatrix} 10 & 0 \\ 0 & 0.001 \end{bmatrix}$, which shows that the speed of convergence in the weakly redundant direction has to be chosen quite smaller than the case of Example 2 where saturation was not present.

For the anti-windup compensator function $k(\cdot)$ we make a naive selection ignoring input saturation and performing an LQR design with identity weights both on the state and the input. The corresponding selection is $k(x_{aw}) = K_{lqr} x_{aw}$ with $K_{lqr} = \begin{bmatrix} 0.944 & 0.0837 & 0.556 \\ -0.244 & 0.203 & 0.15 \end{bmatrix}$. This selection is sufficient to achieve regional stabilization (see, e.g., Teel, Kaiser, and Murray (1997) for an explanation of why any stabilizing gain for (A, B) is already sufficient for regional anti-windup compensation).

The response to the same unit reference used in the previous examples is represented in Fig. 6, where the top plot shows that with these strong saturation constraints there is some slight plant output performance deterioration (solid). Anti-windup without allocation would preserve stability but cause a steady-state performance loss (dashed) due to the steady-state saturation. Finally, no anti-windup and no allocation (dash-dotted) cause

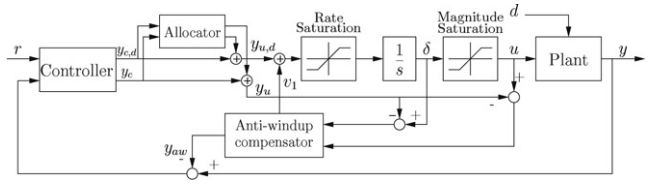

Fig. 7. Dynamic input allocation with anti-windup for magnitude and rate saturated plants.

persistent oscillations. The middle and bottom plots represent the controller output y_c before input allocation and the plant input u after saturation. The plant input converges to the values $[-0.095 \; -0.000337 \; -0.014]$ which means that the percentage of use of the three inputs (as compared to the saturation levels) is roughly [9.5 3.37 7]%. This useful allocation (each input is used by less than 10% of its maximal effort) is automatically enforced by the dynamic allocator. Note that the original allocation performed by the controller (middle plot) is not even within the saturation limits (for example $u_2^* \approx 0.5 \gg m_2 = 0.01$).

5. Nonlinear input allocation for magnitude and rate saturated plants

We address in this section the situation where the plant input is subject to both rate and magnitude saturation. To address rate saturation we will use a similar approach to that one used in Galeani, Onori, Teel, and Zaccarian (2008), which relies on the strong assumption that the controller (4) is strictly proper (namely, $D_c = 0$ and $D_r = 0$). This assumption allows us to naturally extend the dynamic allocation approaches of the previous sections although the main ideas could be used also for nonstrictly proper controllers. Nevertheless, we regard this extension as future work. Due to space constraints, we only discuss the key features of the magnitude + rate saturated case here and refer the interested reader to Zaccarian (2008) for details.

Following a similar trick to that in Galeani et al. (2008), we equip the controller with an extra output corresponding to the derivative of its original output:

$$\begin{aligned} \dot{x}_c &= A_c x_c + B_c u_c + B_r r \\ y_u &= C_c x_c + B_\perp w, \\ y_{u,d} &= C_c A_c x_c + C_c B_c u_c + C_c B_r r + B_\perp \dot{w}, \end{aligned} \tag{15}$$

where $\dot{w}$ denotes the right hand side of the dynamic allocator equation, introduced next. With rate saturation, the dynamic allocator generalizes to the following extended version of (10)[4]:

$$\begin{aligned} \dot{w} &= -KB_\perp^{\mathrm{T}} W(y_u) y_u - K_r B_\perp^{\mathrm{T}} W_r \mathrm{dz}_R(W_r y_{u,d}) \\ y_u &= y_c + B_\perp w \\ W(y_u) &= (\mathrm{diag}((1+\epsilon)M - \mathrm{abs}(\mathrm{sat}_M(y_u))))^{-1}, \end{aligned} \tag{16}$$

where $R = [r_1 \; \cdots \; r_{n_u}]^{\mathrm{T}}$ represents the rate saturation limits of each input channel and $\mathrm{dz}_R(v) := v - \mathrm{sat}_R(v)$ for all $v \in \mathbb{R}^{n_u}$.

The new parameters W_r and K_r in the control allocator (16) play the same role as K and W for the rate saturation aspects. Since rate saturation allocation is only necessary during the transients, we will assume that only the diagonal terms in K_r corresponding to strongly redundant columns of $B_\perp$ are nonzero.

Also in the presence of rate saturation, the augmented control system requires anti-windup action. This anti-windup action is required to be effective for any nonlinear control system inducing closed-loop asymptotic stability without saturation. To this end, we use the novel approach described in Zaccarian (2008) and Forni et al. (2009), which corresponds to a nontrivial extension of the $\mathcal{L}_2$ anti-windup solution of Teel and Kapoor (1997) and Zaccarian and Teel (2002), used in Section 4. Due to space constraints, we do not describe here all the details of the anti-windup implementation, but only summarize briefly the properties of the compensated closed loop. The overall system corresponds to the block diagram of Fig. 7, where the "Anti-windup compensator" block embeds the following dynamics:

$$\begin{aligned} \dot{x}_{aw} &= Ax_{aw} + B(u - y_u) \\ \dot{\delta} &= \mathrm{sat}_R(y_{u,d} + v_1) \\ y_{aw} &= Cx_{aw} + D(u - y_u) \\ u &= \mathrm{sat}_M(\delta), \qquad v_1 = k_r\left(\begin{bmatrix} x_{aw} \\ \delta - y_u \end{bmatrix}\right) \end{aligned} \tag{17}$$

which is interconnected to the controller input via Eq. (12) and where $k_r(\cdot)$ is a stabilizing function generalizing the function $k(\cdot)$ of (11).

The following formal statements (for their proofs see Zaccarian (2008)) generalize the results of the magnitude saturated case (Section 4) to the magnitude + rate saturated case.

Lemma 1. *If $K_r = K_r^{\mathrm{T}} \geq 0$ and $W_r = W_r^{\mathrm{T}}$, then the algebraic loop arising from (16) and the third equation of (15) is well posed.*

Definition 3. A function $y_u(t), t \geq 0$ for which there exists $y_{u,d}(t), t \geq 0$ such that $y_{u,d}(t) = \dot{y}_u(t)$ almost everywhere, is a *feasible control signal* for

$$\begin{aligned} \dot{x}_{aw} &= Ax_{aw} + B(\mathrm{sat}_M(\delta_{aw} + y_u) - y_u) \\ \dot{\delta}_{aw} &= \mathrm{sat}_R\left(y_{u,d} + k_r\left(\begin{bmatrix} x_{aw} \\ \delta_{aw} \end{bmatrix}\right)\right) - y_{u,d} \end{aligned} \tag{18}$$

if $\begin{bmatrix} y_u - \mathrm{sat}_{M-\varepsilon}(y_u) \\ y_{u,d} - \mathrm{sat}_{R-\varepsilon}(y_{u,d}) \end{bmatrix} \in \mathcal{L}_2$ and the solution of (18) satisfies $(x_{aw}, \delta_{aw}) \in \mathcal{L}_2$.

Theorem 4. *Consider an input redundant plant (1) subject to input magnitude saturation of level M, rate saturation of level R and the controller (15) designed in such a way that their interconnection without rate or magnitude saturation and without allocation (namely, $u = C_c x_c$, $u_c = y$) is internally stable. Assume that $B_\perp = [B_{s\perp} B_{o\perp}]$ where the first columns generate the strongly redundant directions and the remaining ones complete the basis of the weakly redundant directions and pick $K_r = K_r^{\mathrm{T}} = \begin{bmatrix} K_{rs} & 0 \\ 0 & 0 \end{bmatrix} \geq 0$, $W_r \geq 0$ diagonal and $K = K^{\mathrm{T}} = \rho\bar{K} > 0$ where $\rho \in \mathbb{R}$ is a positive scalar. Then the origin of the control system with input allocation and without saturation (1), (15) and (16), $u = y_u$, $u_c = y$ is semiglobally exponentially stable with ρ (namely, for each arbitrarily large ball, there is a small enough ρ such that the ball is inside the basin of attraction of the exponentially stable origin). Moreover, for any initial condition $(x(0), x_c(0), w(0))$ in the domain of attraction and any input selections $r(\cdot), d(\cdot)$ such that this system generates a* feasible control signal y_u, *the following holds for the closed loop with input allocation and with saturation (1), (15)–(17) and (12):*

(1) *the plant input response u converges in the $\mathcal{L}_2$ sense to the plant input response of the control system with input allocation and without saturation;*
(2) *if $r(\cdot)$ and $d(\cdot)$ are converging signals, the plant output response y converges in the $\mathcal{L}_2$ sense to the plant output response of the control system without saturation and without allocation.*

Example 4. Consider the same system used in Example 3 and introduce a rate saturation level at all inputs given by $R = [0.3 \; 10 \; 1]^{\mathrm{T}}$, so that, quite reasonably, input 1 which has weak magnitude saturation is characterized by severe rate saturation. For this example we select the same parameters as in Example 3 and use $K_r = \begin{bmatrix} 1000 & 0 \\ 0 & 0 \end{bmatrix}$ and $W_r = I$, so that only the state

[4] Note that in the absence of rate saturation, namely when $R = +\infty$, the dynamic allocator (16) reduces to (10).

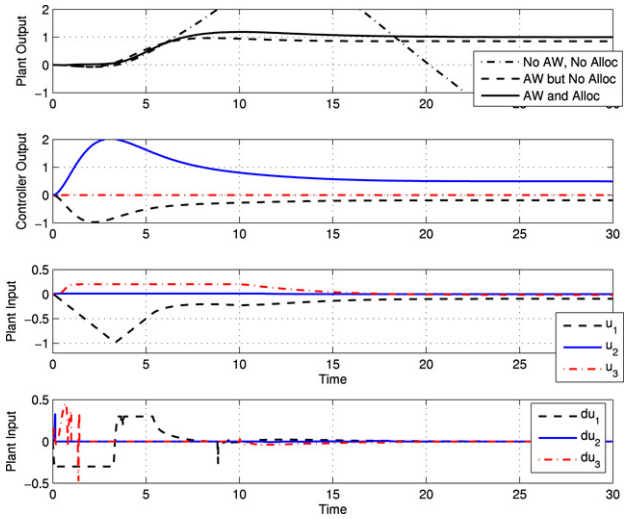

Fig. 8. Example 4: Response to the unit step reference.

redundant directions are used for input rate allocation. Similarly to the magnitude saturated case, the anti-windup gain is selected as $k_r\left(\begin{bmatrix} x_{aw} \\ \delta_{aw} \end{bmatrix}\right) = K_{lqr}\begin{bmatrix} x_{aw} \\ \delta_{aw} \end{bmatrix}$, where K_{lqr} is an LQR gain for system (18) without saturations selected using identity penalty matrices. The closed-loop response to the same unit step reference used in the other examples is represented in Fig. 8. The top and the middle plots mimic the equivalent ones of Fig. 6 with very similar results (instability without AW and steady-state performance loss without allocation). The lower plot shows the plant input derivative which is evidently below the rate saturation limits. Note that in spite of the restrictive rate and magnitude saturation limits, the control allocator with anti-windup compensation can guarantee a desirable transient and converge to a situation with a desirable input allocation.

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